

SAMBP Technical Report TR-78/2028

Adaptive Protection Under N-1 Contingency

Zone Reach Recalculation, Topology Change via GOOSE, Blinder Adjustment: 100-Scenario
IEEE 39-Bus Study

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April 2028 | Chapter 5 (Distance Protection) | Prerequisite: TR-72 (Wide-Area PMU)

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1 Background and Objectives

1.1 Motivation

Distance relay settings are traditionally calculated for the base-case (N-0) network topology. Under N-1 contingency (loss of a line), the network impedances change, and fixed Zone 1/2 settings may over-reach or under-reach, causing miscoordination or missed trips [1]. In IBR-rich networks, N-1 contingencies also change the fault current distribution (infeed changes), compounding the reach error.

The SAMBP adaptive protection framework addresses this by:

1. Detecting topology changes via breaker status GOOSE messages.
2. Recalculating zone reaches from a pre-computed N-1 look-up table.
3. Adjusting blinder settings for load encroachment under heavy power transfer.

1.2 Objectives

1. Implement topology change detection from IEC 61850 GOOSE breaker status.
2. Build an N-1 zone reach table for IEEE 39-bus (39 line outage contingencies).
3. Develop blinder adjustment rules for load encroachment.
4. Validate 100 contingency scenarios: misoperation rate $< 2\%$.

2 Theory and Methodology

2.1 Topology Change Detection

Each circuit breaker publishes its status (OPEN/CLOSE) on the GOOSE bus at ≤ 4 ms latency. The topology change detection algorithm:

1. Monitors the GOOSE dataset for breaker status changes.
2. On detecting a change, updates the network topology model (removes the outaged element from $\mathbf{Y}_{\text{bus}}$).
3. Triggers a zone reach recalculation from the N-1 look-up table.
4. Applies new relay settings within $t_{\text{adapt}} \leq 50$ ms (GOOSE detection: ≤ 4 ms + table lookup: < 1 ms + relay setting write: ≤ 45 ms).

2.2 N-1 Zone Reach Table

For each N-1 contingency c (line outage), the Zone 1 and Zone 2 reaches for relay r are pre-computed:

$$Z_1^{(c)} = 0.80 Z_{\text{line}}^{(c)} / (1 + {}^{(c)}) \quad (1)$$

$$Z_2^{(c)} = 1.20 Z_{\text{line}}^{(c)} + 0.50 Z_{\text{adj}}^{(c)} \quad (2)$$

where $Z_{\text{line}}^{(c)}$ is the protected line impedance (unchanged by the outage, but the source impedance at both ends changes), ${}^{(c)}$ is the IBR current ratio under contingency c (from TR-77 SE), and $Z_{\text{adj}}^{(c)}$ is the adjacent line Zone 1 reach under the contingency.

The table is pre-computed offline for all 39 N-1 contingencies and stored in the SAMBP relay database (total size: $39 \times N_{\text{relays}} \times 4$ parameters).

2.3 Blinder Adjustment for Load Encroachment

Under heavy power transfer (post N-1), the load impedance locus may enter the Zone 2 characteristic. The blinder adjustment restricts the angular reach:

$$\phi_{\text{blinder}}^{(c)} = \phi_{\text{base}} - \Delta\phi \cdot \frac{P_{\text{transfer}}^{(c)}}{P_{\text{rated}}} \quad (3)$$

where $\phi_{\text{base}} = 30^\circ$, $\Delta\phi = 10^\circ$, and $P_{\text{transfer}}^{(c)} / P_{\text{rated}}$ is the normalised post-contingency transfer level from the SE.

3 Simulation Results

3.1 100-Scenario Results

Table 1: TR-78 100-scenario results on IEEE 39-bus.

Contingency type	Scenarios	Correct	Misop.	Misop. rate	Status
Single line outage (light load)	40	40	0	0%	✓
Single line outage (heavy load)	30	29	1	3.3%	✗
Double contingency (N-2)	20	20	0	0%	✓
IBR source outage (N-1 IBR)	10	10	0	0%	✓
Total	100	99	1	1.0%	PASS (< 2%)

Remark 1. *The one misoperation is a heavy-load scenario where the blinder adjustment is insufficient: the blinder angle (20°) still admits a load impedance locus entry for $\cos \phi = 0.85$ lagging at 95% transfer. Resolution: reduce $\Delta\phi$ to 15° for extreme load angles, applied in v1.1 of the N-1 table.*

3.2 Adaptation Time Performance

Table 2: Zone reach adaptation time after N-1 event.

Component	Time (ms)
GOOSE breaker status detection	≤ 4
N-1 table lookup	< 1
New reach parameter write	≤ 45
Total adaptation time	≤ 50
Zone 2 time delay (typical)	400

The 50 ms adaptation time is $8\times$ faster than the Zone 2 time delay (400 ms typical), ensuring that the correct Zone 2 reach is active before any Zone 2 trip decision under the N-1 topology.

4 Conclusion

Finding 1 (N-1 adaptive protection: 1% misoperation rate). *The N-1 adaptive zone reach framework achieves $99/100 = 99\%$ correct operation (1% misoperation), well below the $< 2\%$ criterion. The adaptation time of ≤ 50 ms ensures correct settings before any Zone 2 decision under the new topology.*

Finding 2 (GOOSE topology detection enables adaptive settings). *The IEC 61850 GOOSE breaker status detection provides topology change notification within 4 ms, enabling the SAMBP framework to recalculate zone reaches 50 ms after the contingency — a fundamental improvement over manual periodic setting group changes.*

References

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